DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

VIDYA VIKAS INSTITUTE OF ENGINEERING AND TECHNOLOGY

MYSURU – 570 028

INDIAN KNOWLEDGE SYSTEMS [BIKS609]

ACTIVITY REPORT ON

Foundations of physics in Vaisesika

Submitted by

Deekshith NR 4VM23EC008

Sandeep TR 4VM23EC038

Course Coordinator

Prof. Namratha D’cruz

Assistant Professor

Dept. of ECE, VVIET

Activity Marks

Sl No	Description	Max Marks	Marks Awarded
1	Case Study	10	
2	Study & Consolidation	10	
3	Q & A	10	
4	Presentation	10	
Total Marks		40	
Signature of the Course Coordinator			

2025-2026

Foundations of physics in Vaisesika

INTRODUCTION

The Vaiśeṣika system is one of the six classical schools of Indian philosophy, founded by the sage Kaṇāda. It presents a remarkably structured and logical view of the physical universe, making it highly relevant to what we today call the foundations of physics. Long before modern scientific methods developed, Vaiśeṣika attempted to explain the nature of reality, matter, motion, space, and time using systematic reasoning and categorization.

At its core, Vaiśeṣika is concerned with understanding the nature of existence through a detailed classification of all that exists. It introduces the concept of padārthas (categories of reality), which include substance (dravya), quality (guṇa), motion (karma), universals (sāmānya), particularity (viśeṣa), and inherence (samavāya). Later, a seventh category, non-existence (abhāva), was also included. These categories form the backbone of its physical and metaphysical explanation of the universe.

From a physics perspective, the most important contribution of Vaiśeṣika is its atomic theory. It proposes that all physical substances are composed of indivisible particles called paramāṇu (atoms). These atoms are eternal, indestructible, and combine in different ways to form complex matter. This idea is strikingly similar to the atomic theory developed in modern science, although it emerged independently and through philosophical reasoning rather than experimental evidence.

Vaiśeṣika also explores the concepts of space (ākāśa), time (kāla), and direction (dik) as fundamental entities that enable physical interactions. Unlike modern physics, where space and time are often treated as interconnected (as in relativity), Vaiśeṣika treats them as distinct and independent realities. However, the recognition that such abstract entities are necessary for explaining motion and change shows a deep intuitive understanding of the structure of the universe.

Another significant aspect is the theory of causation. Vaiśeṣika emphasizes that every effect has a cause and classifies causes into different types, such as inherent cause (samavāyi kāraṇa), non-inherent cause (asamavāyi kāraṇa), and efficient cause (nimitta kāraṇa). This classification parallels modern discussions in physics regarding causality and interactions between particles and forces.

he speaker in the video approaches Vaiśeṣika not merely as a philosophical doctrine but as an early scientific framework that attempts to answer fundamental questions about the physical world. The central idea conveyed is that Vaiśeṣika provides a conceptual foundation for physics by identifying the building blocks of reality and explaining how they interact.

According to the speaker, one of the most fascinating aspects of Vaiśeṣika is its atomistic realism. The speaker highlights how the ancient thinkers envisioned atoms as the smallest units of matter that cannot be further divided. These atoms combine in pairs and larger groups to form visible substances. The explanation of material diversity through combinations of atoms is presented as a precursor to modern chemical and physical theories

he speaker further notes that while Vaiśeṣika does not involve mathematical formulations or experimental validation like modern physics, its conceptual clarity and categorization are highly sophisticated. It sets a foundation for thinking about matter, causality, and the structure of the universe in a disciplined way.

In conclusion, the speaker views Vaiśeṣika as a proto-scientific system—a bridge between philosophy and physics. It demonstrates that the quest to understand the universe is not limited to modern times but has deep roots in ancient intellectual traditions. The system's emphasis on atoms, causation, and classification shows that many ideas central to physics today were intuitively explored centuries ago in Indian philosophy.

QUESTION AND ANSWERS

1. Description of the Case

The "case" in the context of Foundations of Physics in Vaiśeṣika can be understood as an intellectual or philosophical debate rather than a legal dispute. It revolves around the attempt to explain the nature of reality, matter, and the physical universe. The central issue is how the universe is structured, what constitutes matter, and how change and motion occur.

The Vaiśeṣika system presents a structured explanation based on atoms (paramāṇu), categories of existence (padārthas), and causation. However, these ideas were often debated by other philosophical schools such as Nyāya, Buddhism, and Vedānta. Thus, the "case" becomes a philosophical argument between different schools of thought, each proposing its own interpretation of reality.

The discussion typically focuses on:

- Whether atoms are the ultimate building blocks of matter
- Whether reality is discrete (atomic) or continuous
- How cause and effect operate in the universe
- The role of perception and inference in understanding reality

2. Accusations Made by Opposing Parties

Opposing philosophical schools raised several criticisms (or "accusations") against the Vaiśeṣika theory:

a) Criticism of Atomic Theory

Some schools, especially Buddhist philosophers, argued that the concept of permanent, indivisible atoms is flawed. They claimed that everything in the universe is momentary (kṣaṇika-vāda), meaning nothing is permanent. According to them, atoms cannot be eternal if all things are constantly changing.

b) Challenge to Inherence (Samavāya)

Vaiśeṣika introduces samavāya (inherence) as a relation that connects substances and their qualities. Critics argued that this concept is unnecessary or abstract, questioning how such a relation can exist independently.

c) Questioning Multiple Categories (Padārthas)

Other schools accused Vaiśeṣika of over-classification, stating that dividing reality into many categories complicates understanding instead of simplifying it. They believed fewer principles could explain the universe more effectively.

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3. Arguments Made by Vaiśeṣika (Defense)

In response to these criticisms, proponents of Vaiśeṣika presented strong logical arguments:

a) Defense of Atomic Theory

Vaiśeṣika philosophers argued that division of matter cannot go on infinitely. There must be a smallest indivisible unit, which they called the atom (paramāṇu). Without such a unit, the existence of stable objects would be impossible.

b) Use of Inference (Anumāna)

They justified the existence of atoms through inference. Even if atoms are not directly visible, their presence is necessary to explain observable phenomena such as combination, separation, and transformation of matter.

c) Justification of Padārthas

Vaiśeṣika defended its classification system by stating that it helps in systematic understanding of reality. Each category (substance, quality, motion, etc.) plays a distinct role in explaining the physical world.

d) Explanation of Causation

They argued that every effect must have a cause and classified causes into different types. This structured approach helps explain how complex objects are formed from simpler components.

4. Arguments Made by Opponents (Prosecution)

Opposing schools continued to challenge Vaiśeṣika with alternative viewpoints:

- Buddhist View: Reality is momentary; no permanent atoms exist
- Vedānta View: The ultimate reality is not atoms but a unified consciousness (Brahman)
- Nyāya View: While partially supporting Vaiśeṣika, they debated certain logical aspects and interpretations

They emphasized:

- The importance of perception over inference
- The possibility of continuous matter instead of discrete atoms
- Simpler explanations over multiple categories

CONCLUSION

The study of the Foundations of Physics in the Vaiśeṣika system highlights that the quest to understand the physical universe has deep intellectual roots that go far beyond modern science. One of the most important takeaways is that Vaiśeṣika presents a systematic and logical framework for explaining reality through clearly defined categories (padārthas), demonstrating an early scientific attitude based on classification, observation, and inference.

A key insight from this study is the concept of atomic theory (paramāṇu), which suggests that all matter is composed of indivisible, eternal particles. This idea closely parallels modern physics, showing that ancient thinkers like Kaṇāda had already begun exploring the fundamental structure of matter through rational thought. The emphasis on causation, motion, and the nature of substances further strengthens its relevance as a foundational approach to physics.

Another major takeaway is the integration of philosophy and science. Unlike modern physics, which often separates empirical study from philosophical inquiry, Vaiśeṣika combines both to provide a more holistic understanding of reality. It not only explains how the physical world works but also explores why it exists in a particular way.

Finally, the debates and criticisms from other schools show that knowledge evolves through questioning and intellectual dialogue. Vaiśeṣika's ability to defend its ideas against opposing views reflects its depth and logical strength.

In conclusion, the Vaiśeṣika system can be seen as a proto-scientific foundation of physics, offering timeless insights into the nature of matter, causality, and the structure of the universe. Its key takeaway is that rational thinking, systematic classification, and inquiry into fundamental principles are essential for understanding the physical world, regardless of the time period.

REFERENCES

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